

The Effects of School Proximity and the School Social Index on Housing Prices

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Agenda

1. Motivation, Existing Literature & Research Question

Why school proximity and social indices matter.

2. Methodology & Data

Our empirical model and the data we have used.

3. Results & Main Findings

Presentation and interpretation of our findings.

4. Conclusion & Limitations

Concluding thoughts and limitations of our work.

Motivation, Existing Literature & Research Question

Motivation

- **Personal Interest:** Analyzing the real estate market's reaction to various factors is particularly interesting from an economic standpoint, as it offers insights into market dynamics. Additionally, the topic is personally intriguing, adding an extra layer of motivation to explore it further.
- **Academic and Practical Relevance:** Housing markets are influenced by various social and economic factors, and school quality is also cited as a underestimated determinant of property values, especially in research (Seo and Simons 2009).

Motivation

Understanding the relationship between school distance, school quality and property prices can provide valuable insights for researchers and individuals interested in the real estate market.

Related Literature Background

Several papers have examined the effect of proximity to schools on housing prices:

- **Empirical Strategies:** While hedonic models are the dominant approach to estimating housing price effects, alternative empirical strategies have also been proposed (see Black and Machin 2011).
- **Findings:** Empirical evidence suggests that proximity to schools positively affects housing prices (Rosiers, Lagana, and Theriault 2001; Huang and Hess 2018).

Related Literature Background

- **Other Factors:** Related work also documents capitalization effects of school reputation (Chin and Foong 2006), differences between public and private schools (Sah, Conroy, and Narwold 2016) and school quality (Edusei, Espey, and Lin 2007; Metz 2015).
- **Environment of studies:** Much of the literature focuses on settings with binding school catchment areas and measures school quality primarily through test scores rather than demographic composition.

Research Question

Out contribution: We study a setting **without binding school catchment areas** (NRW, Germany) and assess whether both school proximity and a **demographically based school social index** are capitalized into housing prices.

Research Question

Research Question: How do school proximity and school social indices affect the prices of single-family houses, and do these effects differ between primary and secondary schools in North Rhine–Westphalia?

H1: House prices differ based on the distance to schools

Expected: Negative coefficients on distance variables

H2: Social index moderates the effect of school proximity on house prices

Expected: Better schools leading to stronger price premiums

Methodology & Data

Empirical Framework

We estimate regression-based hedonic models to

- examine how school proximity and school characteristics are capitalized into housing prices and
- whether the impact of school proximity on housing prices differs by the school's social index

Empirical Framework: Hedonic Price Model

$$\log(P_i) = \beta_0 + \beta_1 \text{Dist}_i + X_i' \gamma + \mu_{\ell(i)} + \varepsilon_i$$

- P_i denotes the listing price of house i
- Dist_i measures the distance to the nearest school (primary or secondary, depending on specification)
- X_i is a vector of housing characteristics
- γ is a vector of coefficients corresponding to X_i
- $\mu_{\ell(i)}$ are municipality-specific characteristics
- ε_i captures unobserved factors

Empirical Framework: Distance and Social Index

$$\log(P_i) = \beta_0 + \beta_1 \text{Dist}_i + \beta_2 \text{Index}_i \\ + \beta_3 (\text{Dist}_i \times \text{Index}_i) + X_i' \gamma + \mu_{\ell(i)} + \varepsilon_i$$

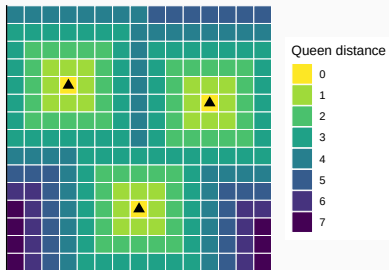
- P_i denotes the listing price of house i
- Dist_i measures the distance to the nearest school (primary or secondary, depending on specification)
- X_i is a vector of housing characteristics
- Index_i is the school social index
- $\mu_{\ell(i)}$ are municipality-specific characteristics
- ε_i captures unobserved factors

Empirical Framework: Queen Contiguity

$$\text{Dist}_{ij}^Q = \max(|x_i - x_j|, |y_i - y_j|)$$

- Results for distance and the social index are re-estimated using Queen (Chebyshev) distance to check robustness of our results.

Queen distance to nearest school on a 15×15 raster grid

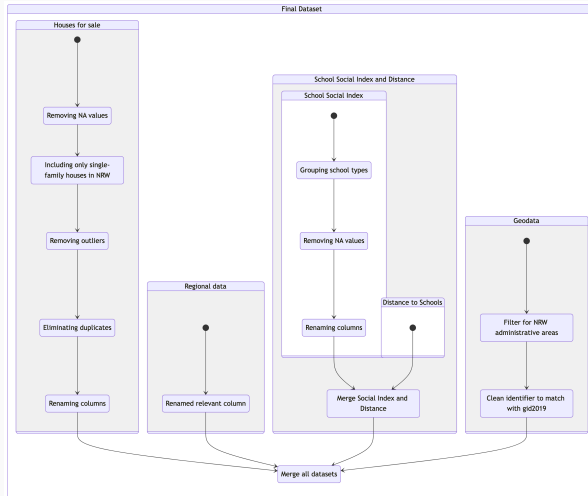


Data used

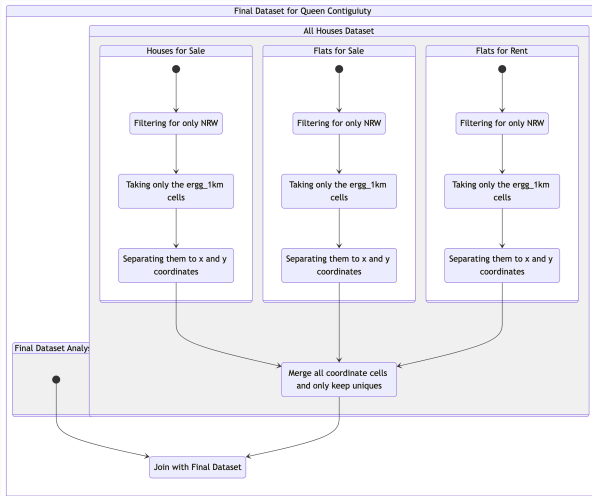
- **Housing Data:** Cross-section real-estate datasets for Germany published by the Research Data Center Ruhr (FDZ Ruhr) (RWI and ImmobilienScout24 2023).
- **School Data:** School social index dataset from North Rhine-Westphalias Ministry of Schools and Education (2024).
- **Regional Data:** Dataset of demographics on municipality-level in NRW. The dataset contains data from INKAR (2020) and Germany's Federal Statistical Office (2011).
- **Geodata:** Geodata for Germany's administrative areas used to create graphics of NRW. Taken from Germany's Federal Agency for Cartography and Geodesy (2025).

Housing and school data both refer to the year 2022.

Pre-Processing of Data: Main Analysis



Pre-Processing of Data: Queen Contiguity



Results and Main Findings

Distance to Schools and House Prices: Hypothesis

H1: House prices differ based on the distance to schools

Expected: Negative coefficients on distance variables

Empirical Framework

- **Outcome:** Log house price
- **Key variables:** Distance to nearest *primary* school, distance to nearest *secondary* school
- **Modeling Strategy:** Continuous, quadratic, and binned specifications

Distance to Schools and House Prices: Regression Strategy

Baseline OLS Controls

Standardizing housing attributes

- Living area
- Plot area
- Number of rooms
- House age

Municipality characteristics

- Population share low income
- Population share middle income

Two Distance Designs

Alternative Specifications

1. Continuous
 - quadratic (km^2)
2. Binned
 - 0–3 km, 3–6 km, 6–9 km, >9 km

Regression Specifications

(1) Naive (Unconditional)

$$\ln(P) = \alpha + \beta_1 D^{prim} + \beta_2 D^{sec} + \varepsilon$$

- **Goal:** Establish raw spatial correlation.
- **Risk:** High Omitted Variable Bias (OVB).

(2) Baseline (Preferred)

$$\ln(P) = \alpha + \mathbf{D} + \mathbf{X} + \varepsilon$$

- **Controls:** Area, Plot, Rooms, Age.
- **Hypothesis:** $\beta < 0$ (Negative semi-elasticity).

(3) Non-Linearity (Polynomial)

$$\ln(P) = \dots + \delta_1 (D^{prim})^2 + \dots$$

- **Test:** $\delta \neq 0$ implies varying MWTP.
- **Logic:** Test for convex/concave decay.

(4–5) Robustness (Single Type)

$$\ln(P) = \alpha + \beta_k D^k + \mathbf{X} + \varepsilon$$

- **Idea:** Estimate primary / secondary separately.
- **Check:** Stability against multicollinearity.

Descriptive Evidence: Price-Distance Gradient

Negative Slope (H1)

Clear decay in prices as distance increases across specifications.

Non-Linearity

Steeper drop in the first 3–6 km, flattening afterwards.

Heterogeneity

Primary schools show a visibly steeper gradient than Secondary.

Fig 1. Continuous Trend (Raw Scatter)

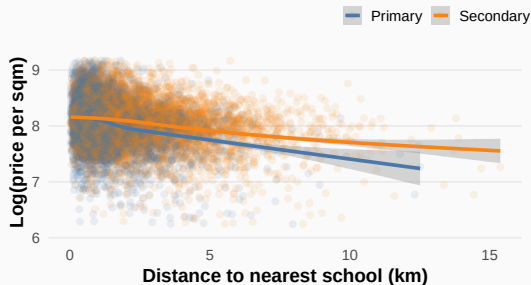
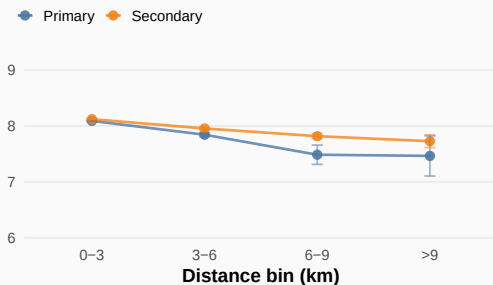


Fig 2. Discrete Gradient (Binned Means)



Main Results: Continuous Model

Variable	Naive		Baseline		Polynomial		Primary Only		Secondary Only	
	Coef.	SE	Coef.	SE	Coef.	SE	Coef.	SE	Coef.	SE
Distance to primary (km)	-0.07***	(0.007)	-0.025***	(0.005)	-0.11***	(0.011)	-0.101***	(0.004)		
Distance to secondary (km)	-0.031***	(0.004)	-0.001	(0.003)	-0.029***	(0.008)			-0.056***	(0.003)
Distance to primary (km) ²					0.008***	(0.002)				
Distance to secondary (km) ²					-0.001	(0.001)				
N	5780		5780		5780		5780		5780	
R ²	0.06		0.577		0.456		0.443		0.437	
Adj. R ²	0.06		0.576		0.455		0.443		0.436	

Note:

Dependent variable: log house price.

HC1 robust standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Main Results: Binned Model

Variable	Naive		Baseline		Primary Only		Secondary Only	
	Coef.	SE	Coef.	SE	Coef.	SE	Coef.	SE
Primary: 3-6 km	-0.145***	(0.027)	-0.142***	(0.021)	-0.259***	(0.019)		
Primary: 6-9 km	-0.36***	(0.094)	-0.355***	(0.072)	-0.543***	(0.07)		
Primary: >9 km	-0.832***	(0.254)	-0.498**	(0.196)	-0.712***	(0.189)		
Secondary: 3-6 km	-0.143***	(0.017)	-0.151***	(0.013)			-0.179***	(0.012)
Secondary: 6-9 km	-0.242***	(0.033)	-0.24***	(0.026)			-0.332***	(0.024)
Secondary: >9 km	-0.21***	(0.08)	-0.259***	(0.062)			-0.435***	(0.059)
N	5780		5780		5780		5780	
R ²	0.046		0.438		0.42		0.413	
Adj. R ²	0.045		0.436		0.419		0.412	

Note:

Dependent variable: log house price. Reference: 0-3 km.

HC1 robust standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Compare: Continuous vs Binned

Variable	Continuous		Binned	
	Coef.	SE	Coef.	SE
Distance to primary (km)	-0.025***	(0.005)		
Distance to secondary (km)	-0.001	(0.003)		
Primary: 3-6 km (ref: 0-3)			-0.142***	(0.021)
Primary: 6-9 km (ref: 0-3)			-0.355***	(0.072)
Primary: >9 km (ref: 0-3)			-0.498**	(0.196)
Secondary: 3-6 km (ref: 0-3)			-0.151***	(0.013)
Secondary: 6-9 km (ref: 0-3)			-0.24***	(0.026)
Secondary: >9 km (ref: 0-3)			-0.259***	(0.062)
N	5780		5780	
R ²	0.577		0.438	
Adj. R ²	0.576		0.436	

Model comparison (main models)

- Continuous fits slightly better.

Model	N	R^2	Adj. R^2	AIC	BIC	CV RMSE	CV MAE	CV R^2
Baseline (Continuous)	5780	0.577	0.576	3304.8	3378.0	0.372	0.289	0.433
Baseline (Binned)	5780	0.438	0.436	4956.5	5049.7	0.377	0.293	0.416

Note:

In-sample: higher R^2 /Adj. R^2 is better; lower AIC/BIC is better. Out-of-sample (5-fold CV): lower RMSE/MAE is better; higher R^2 is better.

Robustness checks (main models)

- Binned is slightly more robust to influential points.

Model	BP p-value	Max VIF	Influential ($D > 4/n$)	Max Cook's D
Baseline (Continuous)	<0.001	1.91	385	0.071
Baseline (Binned)	<0.001	3.00	290	0.030

Note:

Inference uses textbfHC1 robust SE. Lower BP p-values indicate heteroskedasticity; higher VIF indicates collinearity; larger Cook's D indicates influential points.

Robustness check: Queen Contiguity (Distance)

Variable	Primary Schools		Secondary Schools	
	Coefficient	Standard Error	Coefficient	Standard Error
Queen distance	-0.082***	(0.006)	-0.032***	(0.003)

Note:

Dependent variable: log house price, control variables included but not shown.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Economic Mechanisms & Interpretations

1. Capitalization

Why closer = expensive?

- **Convenience Premium:**
Time savings →
Capitalized into property values.
- **Sorting:**
Families cluster → Higher local demand.

2. Primary vs Secondary

Why primary is stronger?

- **Age Constraints:**
Young kids need escorting (Inelastic demand).
- **Micro-Neighborhoods:**
Primary schools anchor local community ties.

3. Non-Linearity

Why gradient flattens?

- **Walkability Limit:**
Premium vanishes after 3km.
- **Substitution:**
Driving → Public transit at longer distances.

Exploring the Social Index Impact on House Prices

H2: Social index moderates the effect of school proximity on house prices

Expected: Better schools leading to stronger price premiums

- **Key Finding from Distance Analysis:** We've already established that *households are willing to pay a premium for proximity to schools*
- **Next Question:** Now, we investigate *whether this price premium is driven by school quality, measured through the Social Index*

What is the Social Index?

Definition: The Social Index is a measure that shows the socio-economic and demographic background of a schools students.

Components of the social index are:

- Parental Education
- Socio-economic Status of the Family
- Migration Background
- Single-Parent Household
- Unemployment in the Family
- Housing Situation

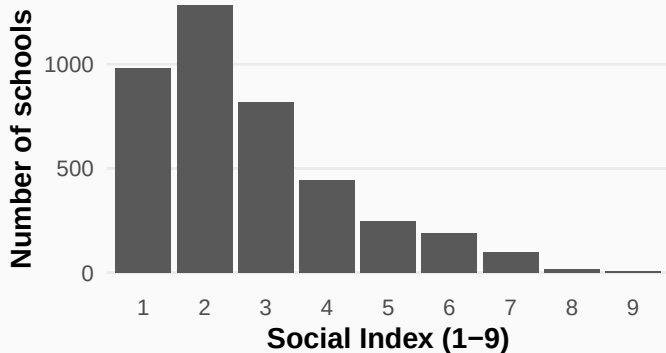
→ It provides insights into the **social environment** that might influence housing prices in that area.

Categorization of the Social Index

Social Index Classification:

- **Good Schools:** Social Index 1-2
- **Average Schools:** Social Index 3-4
- **Bad Schools:** Social Index 5-8
- **Skewed Distribution:** The Social Index classes are heavily skewed, with many schools falling into classes 1, 2 & 3. We excluded class 9 for our analysis because of too less observations

Categorization of the Social Index: Bar Chart

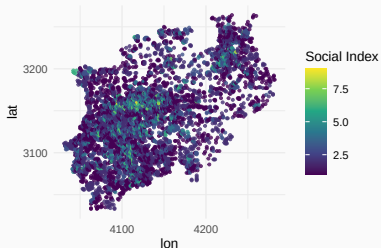


Graph: Distribution of Social Index values across schools showing the skewed nature of the data.

Spatial Distribution of School Social Index in NRW

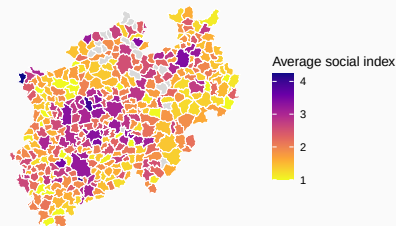
Spatial Distribution of School Social Index in NRW

Each grid cell colored by the social index of the nearest school



Average Social Index by Municipality

Mean household social index within municipalities



Hedonic Regression Model for Social Index

$$\begin{aligned}\log(\text{price}_i) = & \beta_0 + \beta_1 \text{dist_primary}_i + \beta_2 \text{dist_secondary}_i \\ & + \beta_3 \text{school_quality}_i + \beta_4 (\text{dist_primary}_i \times \text{school_quality}_i) \\ & + \beta_5 (\text{dist_secondary}_i \times \text{school_quality}_i) + X_i' \gamma + \mu_{\ell(i)} + \varepsilon_i\end{aligned}$$

- $\log(P_i)$ denotes the natural logarithm of the housing price of house i
- $\text{dist}_i^{\text{primary}}$ measures distance to the nearest primary school (km)
- $\text{dist}_i^{\text{secondary}}$ measures distance to the nearest secondary school (km)
- school_quality_i is a categorical indicator of school quality
- X_i is a vector of housing characteristics
- $\mu_{\ell(i)}$ are municipality-specific characteristics
- ε_i captures unobserved factors

Social Index: Regression Results

Variable	Coefficient	Standard Error
Distance to primary school (km)	-0.08***	(0.007)
School quality: Average	-0.015	(0.021)
School quality: Bad	-0.027	(0.037)
Distance to secondary school (km)	-0.042***	(0.005)
Distance × Average school quality (Primary)	-0.036**	(0.017)
Distance × Bad school quality (Primary)	0.057	(0.046)
Distance × Average school quality (Secondary)	0.035***	(0.013)
Distance × Bad school quality (Secondary)	-0.006	(0.035)

Note:

Dependent variable: log house price.

Reference category for school quality: Good.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robustness check: Social Index

Model	BP p-value	Max VIF	Influential ($D > 4/n$)	Max Cook's D
Good Reference	<0.001	27.57	315	0.033
Average Reference	<0.001	109.13	315	0.033
Bad Reference	<0.001	578.70	315	0.033

Note:

Inference uses textbfHC1 robust SE. Lower BP p-values indicate heteroskedasticity; higher VIF indicates collinearity; larger Cook's D indicates influential points.

Robustness check: Social Index

Variable	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$	GVIF_adj	variable
dist_primary_km	1.793	1	1.339	1.339	Distance to primary school (km)
school_quality	11.667	2	1.848	1.848	School Quality
dist_secondary_km	1.832	1	1.353	1.353	Distance to secondary school (km)
log_area	1.906	1	1.380	1.380	Log building area
log_plot_area	1.233	1	1.110	1.110	Log plot area
zimmeranzahl	1.850	1	1.360	1.360	Number of rooms
house_age	1.102	1	1.050	1.050	House age
dist_primary_km:school_quality	19.146	2	2.092	2.092	Dist primary $\times$ School quality
school_quality:dist_secondary_km	27.570	2	2.291	2.291	School quality $\times$ Dist secondary

Robustness check: Queen Contiguity (Social Index)

Variable	No interaction (Primary)	No interaction (Secondary)	With interaction (Primary)	With interaction (Secondary)
Social index	0.005	0.008*	0.004	0.014**
Social index $\times$ Queen distance	NA	NA	0.002	-0.004

Note: Dependent variable: log house price. Control variables included but not shown. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Social Index: Interpretation of the Results

Primary School: Proximity and School Quality

- **Proximity Effect:** A strong proximity effect persists. Each additional kilometer to the nearest primary school reduces log house prices by 8% (-0.08).
- **School Quality Effect:** There is no robust level effect for school quality. The coefficients for “Average” (-0.015) and “Bad” (-0.027) are negative but not significant, showing no evidence of a “good-school level premium”.

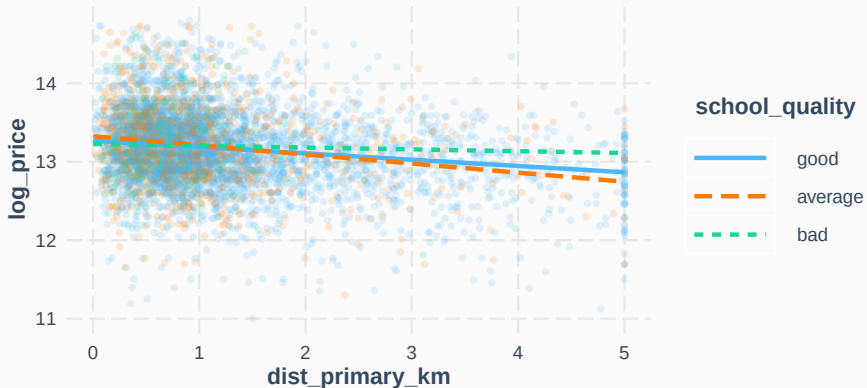
Social Index: Interpretation of the Results

Primary School: Interaction Effect and Bad Schools

- **Interaction Effect:** Proximity is more strongly capitalized for Average schools (interaction -0.036). This contradicts the expectation that “Good” schools should generate the strongest price premium regarding proximity.
- **Bad Schools:** The interaction for Bad schools ($+0.057$) is not significant, possibly due to the limited sample size, indicating no clear difference from “Good” schools.

Social Index: Interpretation of the Results

Simple Slopes Analysis for 'dist_primary_km' by School Quality



Social Index: Interpretation of the Results

Secondary Schools: Proximity and School Quality

- **Proximity Effect:** We observe a similar strong proximity effect for secondary schools. Each additional kilometer to the nearest secondary school reduces log house prices by 4.2% (-0,042).
- **School Quality Effect:** As with primary schools, there is no robust level effect for school quality.

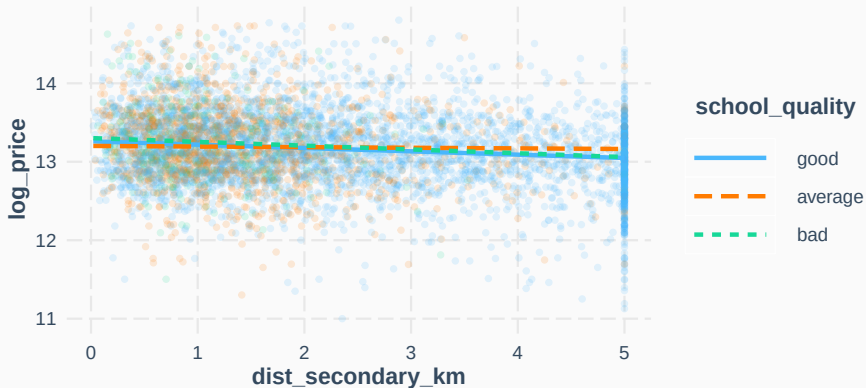
Social Index: Interpretation of the Results

Secondary Schools: Interaction Effect and Bad Schools

- **Interaction Effect:** Proximity is less strongly capitalized for Average secondary schools (interaction $+0.035$), aligning with the expectation that “Good” schools should have the strongest proximity premium.
- **Bad Schools:** The interaction for Bad schools (-0.006) is not significant, possibly due to a limited number of observations.

Social Index: Interpretation of the Results

Simple Slopes Analysis for 'dist_secondary_km' by School Quality



Concluding Thoughts and Limitations

Concluding thoughts: Hypotheses Assessment

H1: House prices differ based on the distance to schools

Expected: Negative coefficients on distance variables

Supported

H2: Social index moderates the effect of school proximity on house prices

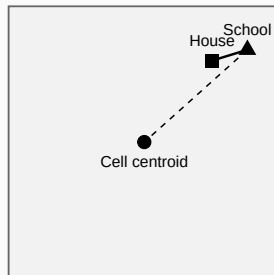
Expected: Better schools leading to stronger price premiums

Not supported

Limitation: Direct Distance

The distance is not measured from the houses to the nearest school directly but from the school to the center of the raster cell.

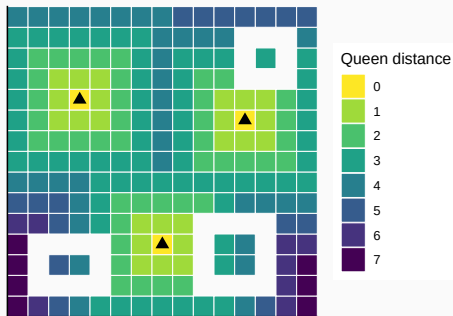
Single raster cell used for distance computation



Limitation: Fragmented Grid

We do not have data for every grid cell of NRW but only a fragmented set of grids containing houses from the available datasets.

Queen distance to nearest school on a 15×15 raster grid



Limitations: Other

- **Omitted neighborhood characteristics:** Unobserved neighborhood attributes correlated with school location may bias estimated school effects.
- **Listing prices:** The prices we have are not transaction but listing prices.
- **Heteroskedasticity:** Despite using robust standard errors, the Breusch-Pagan test indicates the presence of heteroskedasticity, which may still affect the precision of coefficient estimates and standard errors.

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